# ML PS2

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## 1 Probelm 1

1. *Proof.* $e^{2t}y' + 2e^{2t}y = e^{2t}(2 - e^{-4t})$
   $\frac{d(e^{2t}y)}{dt} = 2e^{2t} - e^{-2t}$
   $\int \frac{d(e^{2t}y)}{dt}\, dt = \int (2e^{2t} - e^{-2t})\, dt$
   $e^{2t}y = e^{2t} + \frac{1}{2}e^{-2t} + C$
   $y(t) = 1 + \frac{1}{2}e^{-4t} + Ce^{-2t}$
   Using $y(0) = 1$, $1 = 1 + \frac{1}{2} + C$
   Thus $C = -\frac{1}{2}$.
   Therefore,

$$\boxed{y(t) = 1 - \frac{1}{2}e^{-2t} + \frac{1}{2}e^{-4t}}$$

□

2. $f(t, y) = 2 - e^{-4t} - 2y$ , $y_{n+1} = y_n + hf(t_n, y_n)$

   $t = 0$, $y_1 = y_0 + 2 - 1 - 2y_0$ So, $y_1 = 0$

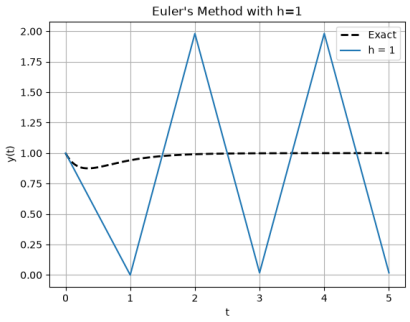
   $t = 1$, $y_2 = y_1 + 2 - e^{-4} - 2y_1$ So, $y_2 = 2 - e^{-4} \approx 1.981684$

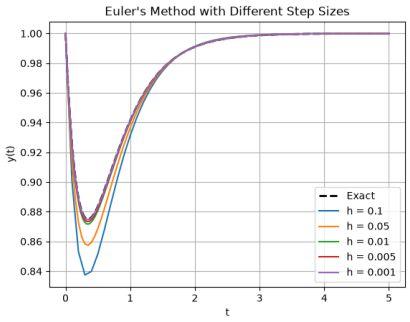
   $t = 2$, $y_3 = y_2 + 2 - e^{-8} - 2y_2$ So, $y_3 = e^{-4} - e^{-8} \approx 0.0180$

   $t = 3$, $y_4 = y_3 + 2 - e^{-12} - 2y_3$ So, $y_4 = 2 - e^{-4} + e^{-8} - e^{-12} \approx 1.9820$

   $t = 4$, $y_5 = y_4 + 2 - e^{-16} - 2y_4$ So, $y_5 = e^{-4} - e^{-8} + e^{-12} - e^{-16} \approx 0.0180$

   $t = 5$, $y_6 = y_5 + 2 - e^{-20} - 2y_5$ So, $y_6 = 2 - e^{-4} + e^{-8} - e^{-12} + e^{-16} - e^{-20} \approx 1.9820$



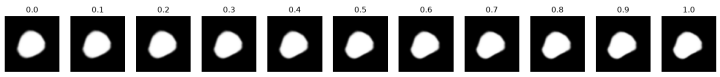

# 2 Probelm 2

1. **Velocity:** [[[-9.4209077e-05 -3.5827211e-04 -5.6035316e-04 ... 5.9041602e-04 4.2646923e-04 1.8404379e-04] [-3.6942988e-04 -6.0255965e-04 -7.0025923e-04 ... 6.1445351e-04 3.0846926e-04 -4.4140299e-05] [-5.4454803e-04 -7.1363128e-04 -7.0386886e-04 ... 5.4073374e-04 1.6213967e-04 -2.2982909e-04] ... [6.5656326e-04 5.2541436e-04 2.3742759e-04 ... 7.8142279e-05 3.9230089e-04 6.0485688e-04] [4.8666575e-04 2.7613185e-04 -3.6954483e-05 ... 2.9674117e-04 4.8171671e-04 5.5559736e-04] [2.1531215e-04 -4.1308595e-05 -3.2184779e-04 ... 4.7715576e-04 4.9033645e-04 4.0145847e-04]]

   [[-4.6146748e-04 -8.7501633e-04 -1.0771136e-03 ... 9.8233926e-04 5.8823067e-04 6.6772351e-05] [1.4740908e-04 -2.1338616e-04 -5.1986641e-04 ... 7.6469936e-04 7.0038391e-04 4.7767465e-04] [7.0743141e-04 4.8590105e-04 1.5182293e-04 ... 3.6554254e-04 6.3984137e-04 7.6379417e-04] ... [-1.1901486e-03 -1.3232563e-03 -1.1420052e-03 ... 4.9742172e-04 -1.6078344e-04 -7.7012851e-04] [-1.2356673e-03 -1.5120856e-03 -1.4278860e-03 ... 8.1333215e-04 7.8863297e-05 -6.6098780e-04] [-9.6670643e-04 -1.3466324e-03 -1.4039861e-03 ... 9.9189184e-04 3.5507281e-04 -3.5211834e-04]]]

   Figure 1 shows the resulting time sequence $s \circ \phi_t$ for

$$t = 0, 0.1, \ldots, 1.0.$$




2. **Running Time**

   The running time for generating the sequence was approximately

   $$\boxed{0.147 \text{ seconds}}$$

3. **Observation** The higher the $\mu$ is, the skinnier the object is (¿0); The smaller the $\mu$ is, the fatter the object is(opposite direction (The edge). The $\sigma$ make a difference in the hole(contrast part).

# 3 Probelm 3

I would say different deformation direction. Problem 2 is squeezing the object, while object in Problem 3 is becoming a round shape(expanding).